

Research Statement

My research focuses on developing scalable and interactive **Multi-Modal Generative AI for Physical AI**. Physical AI encompasses perception, understanding, prediction, and interaction with the physical world by learning the properties and dynamics of real-world environments. It has broad applications in areas such as robotics and autonomous vehicles. Towards it, I have published 6 first-author works (all open-sourced) and have two works selected as Best Paper Award Finalist at CVPR and ICCV. My research can be divided into the following aspects:

1. Embodied Perception: During my undergraduate studies, I completed my bachelor’s thesis on **PanopticNeRF&PanopticNeRF-360**, which leverage neural rendering to generate high-quality 2D panoptic segmentation annotations from a set of sparse images and easy-to-obtain coarse 3D bounding primitives in urban scenes. Beyond this scene-level understanding and data collection, I also serve as a core contributor to **OmniObject3D**, a large-vocabulary 3D object dataset with massive high-quality real-scanned 3D objects. To further advance general geometric understanding, I proposed **GeoWizard**, which is a generative foundation model for estimating depth and normal from a single image, enabling it to capture richer high-frequency geometric details compared to previous discriminative methods.

2. Embodied Simulation: Recent advances in video generative models have lowered the barrier to simulating the spatio-temporal dynamics of objects and scenes. At the object level, I proposed **3DTrajMaster**, a video diffusion model that enables controllable multi-entity motion in 3D space for text-to-video (T2V) generation by leveraging entity-specific 3D pose sequences as additional inputs. I also contributed the 360-Motion Dataset, which associates collected 3D human and animal assets with GPT-generated motion trajectory across diverse 3D UE platforms. At the scene level, I am a core contributor to **SynCamMaster**, which generates synchronized multi-view videos from a single text prompt, and **ReCamMaster**, which re-renders an input video under a novel camera trajectory. To further equip ReCamMaster with long-term spatio-temporal memory, I proposed **PlenopticDreamer**, an autoregressive architecture with a 3D FOV-based video retrieval mechanism for scalable and coherent multi-camera generation. I also introduced progressive context-scaling and self-conditioned training strategies to enhance stability and long-term consistency. PlenopticDreamer has been integrated into **NVIDIA Cosmos2.5 product**, supporting downstream application teams such as the Metropolis and Automotive teams. Under **NVIDIA Cosmos-Transfer2.5**, it transforms third-person robotic manipulation video to synchronized multi-view videos, and converts head-view robotic manipulation video to synchronized hand-view multi-view videos. Building on top of it, I further contributed to the NVIDIA Multi-View Robotic Transfer project, which converts multi-view edge/segmentation/depth/blue videos to realistic and high-fidelity videos. Under **NVIDIA Cosmos-Predict2.5**, it generates synchronized multi-view robotic manipulation videos from multi-view images and camera trajectories, enabling policy training for robots. The system has been deployed to downstream partners, including Samsung and XPeng. To support real-world, low-latency applications, I enhanced NVIDIA Cosmos2.5 with interactive capabilities through post-training efforts, including data curation, model distillation, and self-forcing training on Transfer2.5. Meanwhile, to further solve the data scarcity of robotic manipulation data, I proposed **RoboMaster**, a paradigm that include: interactive 2D trajectory input $\rightarrow$ I2V robotic manipulation demonstrations $\rightarrow$ actions via inverse dynamic model.

3. Embodied Reasoning: Moving forward, I recognized that physical reasoning is also a crucial component in the loop: from physical perception $\rightarrow$ reasoning and planning $\rightarrow$ simulation and error correction. Currently, I am helping develop **Cosmos3**, a world foundation model designed for end-to-end physical perception, reasoning, simulation, and action. I firmly believe that in the future, an OmniModel that integrates understanding, reasoning, and simulation will have a far greater impact on physical AI, and pursuing this vision is my long-term goal.